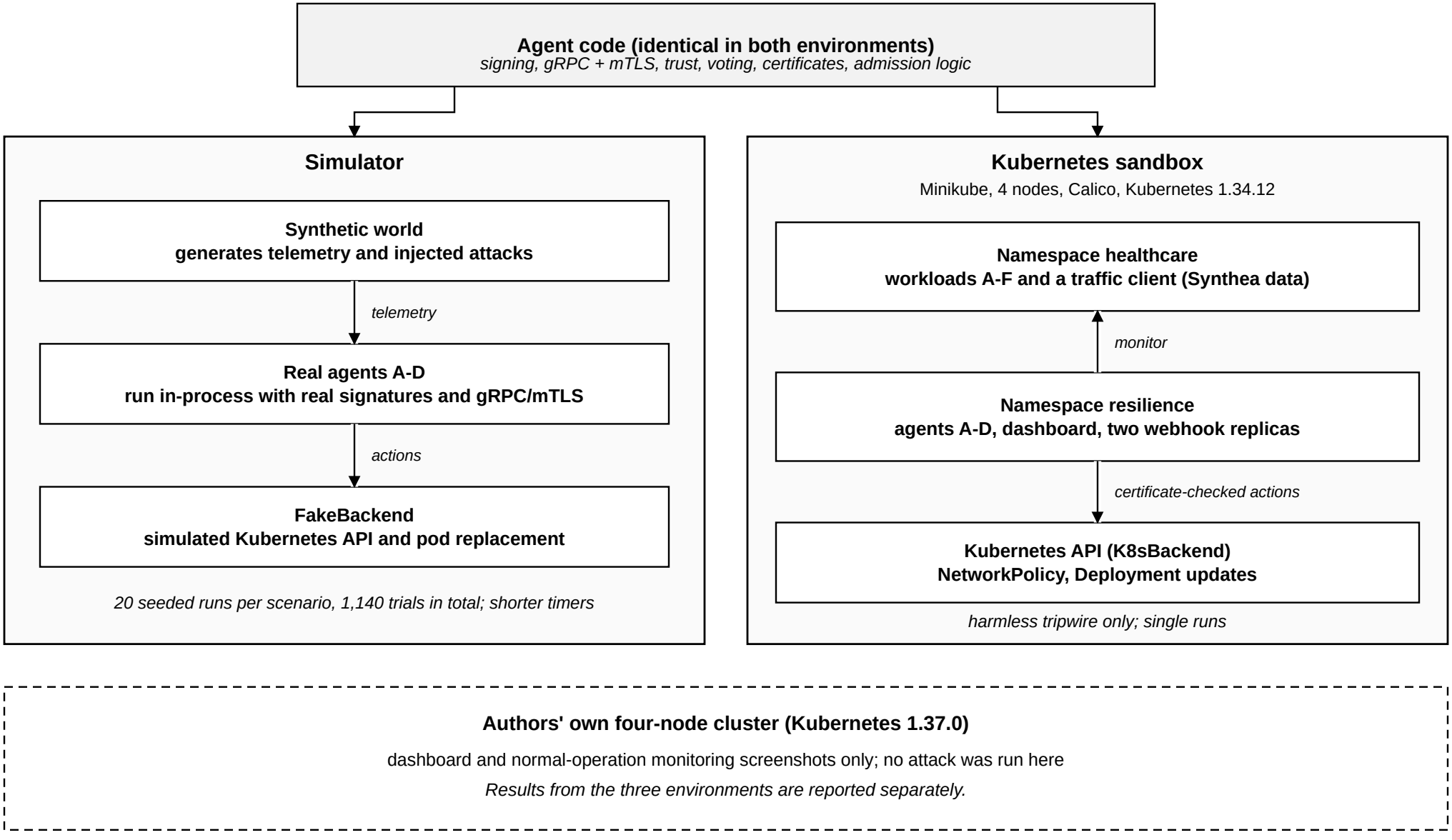
Agent code (identical in both environments)
signing, gRPC + mTLS, trust, voting, certificates, admission logic
Simulator
Synthetic world
generates telemetry and injected attacks
telemetry
Real agents A-D
run in-process with real signatures and gRPC/mTLS
actions
FakeBackend
simulated Kubernetes API and pod replacement
20 seeded runs per scenario, 1,140 trials in total; shorter timers
Kubernetes sandbox
Minikube, 4 nodes, Calico, Kubernetes 1.34.12
Namespace healthcare
workloads A-F and a traffic client (Synthea data)
monitor
Namespace resilience
agents A-D, dashboard, two webhook replicas
certificate-checked actions
Kubernetes API (K8sBackend)
NetworkPolicy, Deployment updates
harmless tripwire only; single runs
Authors' own four-node cluster (Kubernetes 1.37.0)
dashboard and normal-operation monitoring screenshots only; no attack was run here
Results from the three environments are reported separately.